

Section 3.1: Confidence Intervals

Introduction

Up until now, we have been using hypothesis tests to determine if a value for a parameter is plausible. Our new goal is **estimation**: what values of the parameter are plausible?

Confidence Interval: a range of likely values for the parameter of interest.

Example: Legalization of Medical Marijuana

In February of 2014, a Star Tribune Minnesota poll asked a random sample of adults in Minnesota, “Do you support or oppose legalizing marijuana for medical purposes in Minnesota?” Of the 800 adults surveyed, 408 (51%) answered that they would support this.

Ultimately, the Star Tribune conducted this poll because they wanted to estimate the proportion of adults in Minnesota that felt (at the time of the survey) that medical marijuana should be legalized. Keep this in mind as you answer the following questions.

Source: <http://www.startribune.com/politics/statelocal/245910931.html>

Answer the following questions:

- 1) What is the population of interest?

All adults in Minnesota

- 2) What is the parameter of interest?

Data is categorical

The long run proportion of adults in Minnesota who support legalizing medical marijuana in Minnesota

- 3) What is the sample?

800 adults in MN that were surveyed

- 4) What is the observed statistic of interest? This is also called a **point estimation**.

$408/800 = 0.51$

- 5) What would happen if you took a different sample of 800 Minnesota adults?

Might see 49% approve, or 52% approve. The point estimate will change depending on the sample.

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Hypothesis

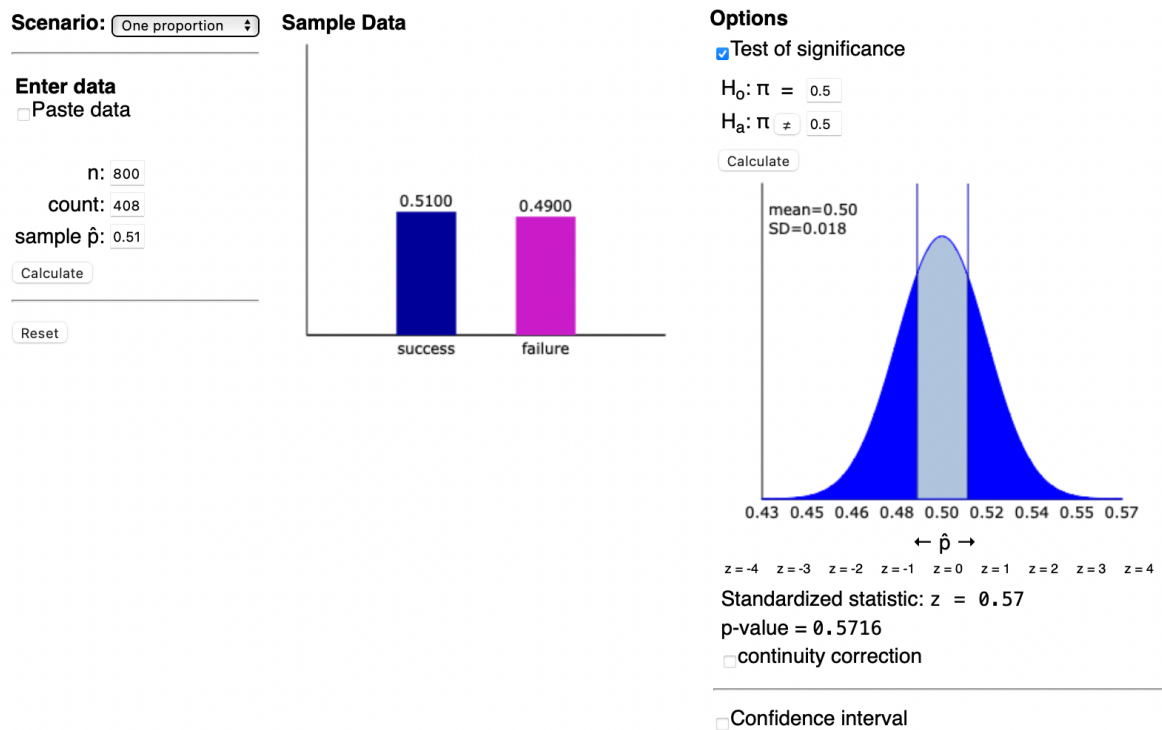
Null: The long run proportion of Minnesotans who feel medical marijuana should be legalized is equal to 0.5

In symbols: $H_0: \pi = 0.5$

Alternative: The long run proportion of Minnesotans who feel medical marijuana should be legalized is different from 0.5

In symbols: $H_A: \pi \neq 0.5$

Calculate the p-value from the **Theory Based Inference Applet**.



Note: when setting up the applet, you either enter the number of successes into the “count” box or the point estimation into “sample $\hat{p}$ ”. The other box will automatically fill in.

- 6) Based off the p-value, would you Reject (R) or Fail to Reject (FTR) at a significance level of 0.05?

FTR because p-value = 0.5716 > 0.05

The general idea behind constructing a confidence interval is that we can use our sample to provide a plausible range of parameter values.

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Using the Theory Based Inference applet, change the values of π to fill in the following table.
(i.e., change the hypothesis test for different parameter values while using the same sample)

π	0.45	0.46	0.47	0.48	0.49	0.50	0.51	0.52	0.53	0.54	0.55
p-value	0.0006	0.0045	0.0234	0.0894	0.2578	0.5716	1	0.5713	0.2570	0.0887	0.0230
R/FTR	R	R	R	FTR	FTR	FTR	FTR	FTR	FTR	FTR	R

7) What trends do we see?

R ----- (FTR) ----- R

8) When we Fail to Reject, are we saying the null or alternative is plausible?

The null is plausible

9) What is the interval of plausible values?

0.48 to 0.54

Interpretation: We are 95% confident that the long-run proportion of Minnesotans who feel that medical marijuana should be legalized is between 0.48 and 0.54.

General Form of a Confidence Interval

Notice how the confidence interval is centered around the point estimate.

$$\text{Confidence Interval (CI)} = \text{Point Estimate} \pm \text{Margin of Error}$$

In the previous example, the confidence interval can be expressed as:

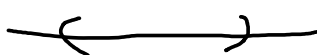
$$51\% \pm 3.5\% \text{ (i.e., } 47.5\% \leq \pi \leq 54.5\%).$$

Significance Levels (α)

Significance Level + Confidence Level = 100%

When $\alpha = 0.10$,

- Relates to a 90% CI
- Rejects when p-value < 0.10
- Narrower than 95% CI
- Rejects the null more often (remember that $\alpha = 0.10$ means that we reject the null 10 percent of the time)

95% 

90% 

When $\alpha = 0.05$,

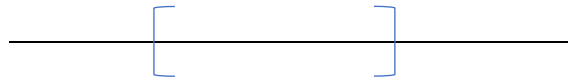
- Relates to a 95% CI
- Rejects when p-value < 0.05

When $\alpha = 0.01$,

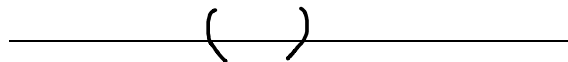
- Relates to a 99% CI
- Rejects when p-value < 0.01
- Wider than 95% CI
- Rejects the null less often than 95% CI

Using the 95% confidence interval as a reference, draw what you would expect from the 90% confidence interval and 99% confidence interval

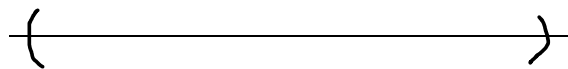
95% CI



90% CI



99% CI



Technical Note About Confidence Intervals

We say that we are “ $100 - \alpha * 100\%$ confident” because $100 - \alpha * 100\%$ of the confidence intervals generated from samples of a given population will cover the true parameter value. In the instance of $\alpha = 0.05$, this means that 95% of confidence intervals generated from samples within a population will cover the true parameter value. In Section 3.3, we will look at an applet that will demonstrate what this means. This is NOT the same as saying that the probability of the parameter being in the confidence interval is $100 - \alpha * 100\%$

Section 3.2: Confidence Intervals for One Proportion

Formula

The general formula for confidence intervals is as follows:

$$\left[\text{Observed Statistic} \pm \text{Multiplier} \cdot \text{Standard Error} \right]$$

Unlike in Chapter 1, we now have a new formula for the standard error:

$$SE = \sqrt{\frac{\hat{p} \cdot (1 - \hat{p})}{n}}$$

$$\hat{p} = 0.51$$
$$SE = \sqrt{\frac{0.51(0.49)}{800}}$$

From Ch 1, $SE = \sqrt{\pi_0 \cdot (1 - \pi_0) / n}$

Notice how we are no longer using π in the calculation, but rather $\hat{p}$.

To construct a 95% Confidence interval, the multiplier we will use in this course is 2. This gives us the confidence interval for one proportions to be:

$$CI = \hat{p} \pm 2 * \sqrt{\frac{\hat{p} * (1 - \hat{p})}{n}} = (\text{Lower Limit}, \text{Upper Limit})$$

Technical aside: if you look into confidence intervals from sources outside of our class, you will notice that the multiplier value for a 95% confidence interval will likely be 1.96. However, we will **only use a multiplier value of 2 for this course.**

Example: Legalization of Medical Marijuana

Recall the medical marijuana survey question from Section 3.1. Calculate the 95% confidence interval.

In February of 2014, a Star Tribune Minnesota poll asked a random sample of adults in Minnesota, "Do you support or oppose legalizing marijuana for medical purposes in Minnesota?" Of the 800 adults surveyed, 408 (51%) answered that they would support this.

$$\hat{p} = 0.51$$

$$n = 800$$

$$SE = \sqrt{\frac{\hat{p} * (1 - \hat{p})}{n}} = 0.01767413$$

$$\hat{p} \pm 2 * SE = 0.51 \pm 2 * 0.01767413 = (0.475, 0.545)$$

Conclusions Based Off of Confidence Intervals

Recall the general interpretation of a confidence interval.

- We are 95% confident that the long run proportion of “context” falls between (*lower limit*) and (*upper limit*).

If the null hypothesis value is inside of the interval

- It is a likely value
- Fail to Reject

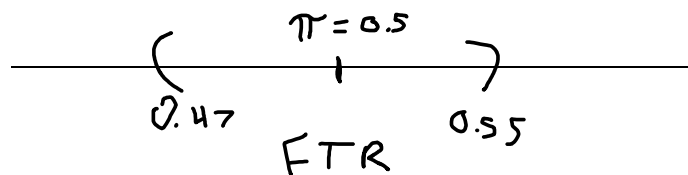
If the null hypothesis value is outside of the interval

- It is not a likely value
- Reject

Conclusion: We (do/do not) have evidence to conclude (restate alternative in context).

In the context of the medical marijuana example,

- Hypothesis
 - $H_0: \pi = 0.5$
 - $H_A: \pi \neq 0.5$
- Confidence Interval
 - $(0.47, 0.55) \rightarrow$ Fail to Reject since 0.5 falls inside our interval



- Interpretation of Confidence Interval
 - We are 95% confident the long run proportion of adults in Minnesota who feel medical marijuana should be legalized falls between 0.47 and 0.55.
- Conclusion in context
 - We do not have evidence to conclude the long run proportion of adults in Minnesota who feel medical marijuana should be legalized is different from 0.5

Conclusions



	P-value	Standardized Statistic	Confidence Interval
Reject	< 0.05	"Outside the 2's"	Hypothesized value outside of the CI
Fail to Reject	> 0.05	"Between the 2's"	Hypothesized value inside of the CI

Section 3.3: Confidence Intervals for Single Means

Formula

The general formula for confidence intervals also applies to single means.

$$\text{Observed Statistic} \pm \text{Multiplier} * \text{Standard Error}$$

Before calculating the confidence interval, be sure that the validity conditions are met for the theory approach (sample size greater than 20 and the distribution is not strongly skewed).

The formula for the standard error is the same as from Chapter 2.

$$SE = \frac{s}{\sqrt{n}}$$

95% Confidence interval

$$CI = \bar{x} \pm 2 * \frac{s}{\sqrt{n}} = (\text{Lower Limit}, \text{Upper Limit})$$

Example: Textbooks

Suppose that two Winona State University students gathered data on prices for a random sample of 30 textbooks from the campus bookstore. They found the average price was \$165.02 and the standard deviation was \$31.42. The data is plotted below.